```

============================================================
  3. LOADING RAW DATA
============================================================
Raw shape  : 5,000 rows × 31 columns
Columns    : ['Employee_ID', 'Age', 'Gender', 'Job_Role', 'Industry',
'Years_of_Experience', 'Work_Location', 'Hours_Worked_Per_Week',
'Number_of_Virtual_Meetings', 'Work_Life_Balance_Rating', 'Stress_Level',
'Mental_Health_Condition', 'Access_to_Mental_Health_Resources', 'Productivity_Change',
'Social_Isolation_Rating', 'Satisfaction_with_Remote_Work',
'Company_Support_for_Remote_Work', 'Physical_Activity', 'Sleep_Quality', 'Region',
'Avg_Device_Watts', 'Hours_Per_Day', 'Daily_kWh', 'Monthly_kWh',
'Electricity_Consumption_kWh', 'Electricity_Cost_EGP', 'Food_Cost', 'Transport_Cost',
'Performance', 'Country', 'Season']

============================================================
  4. DATA CLEANING
============================================================
Duplicate rows found : 0

Missing values (columns with gaps):
   Physical_Activity                          1629 (32.6%)
   Mental_Health_Condition                    1196 (23.9%)
   Number_of_Virtual_Meetings                  327 (6.5%)

Imputation:
   [Number_of_Virtual_Meetings] filled 327 NaN with median = 8.00

Remaining nulls after imputation : 3152

Outlier capping (IQR ×1.5 whiskers):
   Electricity_Consumption_kWh                4 outliers capped  [32.98, 462.43]
   Electricity_Cost_EGP                     625 outliers capped  [-599.35, 3458.71]
   Performance                               44 outliers capped  [66.33, 93.69]

Cleaned data saved → cleaned_data.csv  (shape: (5000, 31))

============================================================
  5. DESCRIPTIVE STATISTICS
============================================================
                                    Mean     Std     Min   Median      Max
Age                                41.00   11.30   22.00    41.00    60.00
Years_of_Experience                17.81   10.02    1.00    18.00    35.00
Hours_Worked_Per_Week              39.61   11.86   20.00    40.00    60.00
Number_of_Virtual_Meetings          8.08    4.18    1.00     8.00    15.00
Work_Life_Balance_Rating            2.98    1.41    1.00     3.00     5.00
Social_Isolation_Rating             2.99    1.39    1.00     3.00     5.00
Company_Support_for_Remote_Work     3.01    1.40    1.00     3.00     5.00
Avg_Device_Watts                  668.56   94.94  600.00   600.00   800.00
Hours_Per_Day                       7.92    2.37    4.00     8.00    12.00
Daily_kWh                           5.29    1.76    2.40     5.16     9.60
Monthly_kWh                       116.48   38.76   52.80   113.52   211.20
Electricity_Consumption_kWh       251.05   70.22   60.56   246.38   462.43
Electricity_Cost_EGP             1547.25  975.29  116.21  1367.93  3458.71
Food_Cost                        1276.31  269.99  590.08  1264.92  2083.77
Transport_Cost                    690.87  431.40    0.00   689.67  1939.11
Performance                        79.99    5.14   66.33    80.01    93.69

============================================================
  6. CORRELATION ANALYSIS
============================================================
Top 10 feature correlations:
                  Feature_A                   Feature_B  Correlation
                  Daily_kWh                 Monthly_kWh     1.000000
      Hours_Worked_Per_Week               Hours_Per_Day     1.000000
      Hours_Worked_Per_Week                 Monthly_kWh     0.895639
              Hours_Per_Day                   Daily_kWh     0.895639
      Hours_Worked_Per_Week                   Daily_kWh     0.895639
              Hours_Per_Day                 Monthly_kWh     0.895639
```

```
            Avg_Device_Watts              Transport_Cost    -0.815526
Electricity_Consumption_kWh              Transport_Cost    -0.766150
            Avg_Device_Watts Electricity_Consumption_kWh     0.710101
        Electricity_Cost_EGP                   Food_Cost     0.432306

============================================================
  7. WORKFORCE DEMOGRAPHICS
============================================================
   Female            1274 (25.5%)
   Male              1270 (25.4%)
   Prefer Not To Say  1242 (24.8%)
   Non-Binary        1214 (24.3%)
   Avg Age              : 41.0 yrs
   Avg Years Experience: 17.8 yrs

============================================================
  8. WORK-LOCATION & REMOTE-WORK ANALYSIS
============================================================
Work_Location
Remote    1714
Hybrid    1649
Onsite    1637

Satisfaction scores by work location:
Work_Location
Onsite    2.03
Hybrid    2.01
Remote    1.96

============================================================
  9. MENTAL HEALTH & STRESS ANALYSIS
============================================================
Mental health conditions:
Mental_Health_Condition
Burnout       1280
Anxiety       1278
Depression    1246

Stress levels:
Stress_Level
High      1686
Medium    1669
Low       1645

Avg performance by stress:
Stress_Level
Low       80.08
Medium    79.98
High      79.92

============================================================
  10. PRODUCTIVITY & PERFORMANCE
============================================================
Productivity change breakdown:
Productivity_Change
Decrease     1737
No Change    1677
Increase     1586

Avg performance score : 79.99
Median performance    : 80.01

============================================================
  11. ENERGY CONSUMPTION & COST ANALYSIS
============================================================
Avg Monthly kWh by Region:
Region
South America    117.91
Asia             117.26
```

```
North America    116.85
Africa           115.91
Europe           115.67
Oceania          115.41

Avg Electricity Cost by Season:
Season
Summer    1569.53
Winter    1525.46

============================================================
  12. WORK-LIFE BALANCE & SLEEP QUALITY
============================================================
Avg Performance by Sleep Quality:
Sleep_Quality
Poor       80.06
Good       80.06
Average    79.85

Correlation (Hours Worked ↔ Work-Life Balance): r = 0.001  (p=0.9538)

============================================================
  13. INDUSTRY & COUNTRY TRENDS
============================================================
Employee count by Industry:
Industry
Finance          747
It               746
Healthcare       728
Retail           726
Education        690
Manufacturing    683
Consulting       680

Avg Performance by Country:
Country
Usa          80.20
Brazil       80.11
Australia    79.99
Egypt        79.98
Uk           79.91
Japan        79.78

============================================================
  14. KEY INSIGHTS SUMMARY
============================================================

1. STRESS & PERFORMANCE: Employees with HIGH stress score 79.92 vs 80.08
for LOW stress (lower by 0.16 pts).

2. REMOTE SATISFACTION: 'Onsite' workers report the highest satisfaction
score (2.03/3).

3. SLEEP & PERFORMANCE: 'Poor' sleepers achieve the best avg performance
(80.06).

4. MENTAL-HEALTH RESOURCES: Access=Yes → avg performance 79.96 vs
Access=No → 80.03.

5. ENERGY: 'South America' has the highest avg monthly consumption
(117.9 kWh/month).

6. PRODUCTIVITY: The most common self-reported change is 'Decrease'
(1737 employees, 34.7%).

7. WORK-LIFE BALANCE: Positive correlation (r=0.001) between hours
worked and work-life balance rating (more hours = higher balance).

8. TOP INDUSTRY: 'Retail' leads in avg performance (80.29).
```